UNIT 2 · TOPIC 2(A)

Bearings

Syllabus topic: Design of bearings, couplings, springs and fasteners

Sliding (plain) bearings

Bush/journal bearings (bronze, babbitt, sintered self-lubricating, wood/lignum on some agricultural rollers) run on a hydrodynamic film. They are simple, damp shock, and suit low speeds and dusty sites.

Rolling-element ("antifriction") bearings

Bearing	Load / use
Deep-groove ball	Radial + some axial; general purpose
Self-aligning ball / spherical roller	Tolerate misalignment — long shafts of farm machines
Cylindrical roller	High radial capacity
Needle roller	Limited radial space — U-joints use needle bearings
Tapered roller	Combined radial + axial — wheel hubs, final drives; mounted in pairs
Thrust	Axial loads

Rating life

$$L_{10} = (C/P)^p \text{ million revolutions } (p = 3 \text{ for ball, } 10/3 \text{ for roller bearings})$$

- **L_{10} (B_{10})** = the life that **90%** of a group of bearings will complete or exceed.
- C = basic dynamic capacity; P = equivalent load.

Agricultural mountings

Flanged/pillow-block housings with triple-lip seals, relubricatable or sealed-for-life. Disc-harrow gang bearings are heavy-duty sealed units.

Practice MCQs

Q28. L_{10} (B_{10}) bearing life is the life that:

(a) 10% of bearings complete

(b) 90% of a group of bearings will complete or exceed

(c) Average bearing achieves

(d) The best bearing achieves

Q29. Bearing life varies with load as:

(a) $L \propto C/P$

(b) $L \propto (C/P)^3$ for ball bearings

(c) $L \propto (C/P)^2$

(d) L ∝ e^(C/P)

Q30. For combined heavy radial and axial loads in tractor wheel hubs, the bearing used is:

(a) Deep-groove ball

(b) Needle roller

(c) Tapered roller

(d) Self-aligning ball

Q31. Universal-joint cross trunnions usually run in:

(a) Bush bearings
(b) Needle roller bearings
(c) Ball bearings
(d) Tapered rollers

Answers

Q28 — (b) 90% of a group of bearings will complete or exceed — *L_{10} = (C/P)^p million rev; 90% reliability definition — repeatedly asked.*

Q29 — (b) L ∝ $(C/P)^3$ for ball bearings — *L_{10} = (C/P)^p million rev, with p = 3 for ball bearings.*

Q30 — (c) Tapered roller

Q31 — (b) Needle roller bearings

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 4 (Elements of Rotary Power-Transmission Systems) and Appendix (ASAE V-belt standards); standard machine design references (Shigley; V.B. Bhandari); ASAE/IS standards.